

MICHAEL LEE

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OBJECTIVE

As a passionate and driven individual with a strong Computer Science and Software Engineering foundation, I am committed to leveraging my technical expertise and innovative mindset to contribute to transformative software projects. I thrive in dynamic environments and am eager to collaborate with teams that push technological boundaries. I aim to apply my skills to create impactful solutions while continuously growing in a setting that values learning and innovation.

EDUCATION

University of Washington

Master of Science in Computer Science & Software Engineering | GPA: 3.67 **Expected June 2025**

- *Relevant Coursework:* Parallel Programming, Multimedia Databases, Evaluating Software Design, Mobile Computing, Software Architecture

Bachelor of Science in Computer Science & Software Engineering | GPA: 3.63 **August 2021**

SOFTWARE ENGINEERING SKILLS

Languages: Java, C++, Python, Swift, SQL, JSON, XML, Dart, Flutter

Database Systems: MySQL, SQLite

Dev Ops and Version Control: Azure, Git

Tools and technologies: AWS EC2, OpenMP, MPI, MapReduce, Hadoop, Spark

SOFTWARE ENGINEERING EXPERIENCE

Luna mHealth Research Project

April 2024 - Present

Research Assistant

- Collaborated with a cross-functional team to develop a mobile health education app for rural populations with limited healthcare access.
- Helped design and implement an authoring system, enabling healthcare professionals to create and distribute custom health education modules.
- Led the team in rendering and optimizing UI elements for performance and ensuring a smooth user experience across devices.
- Participated in Agile development cycles, ensuring timely feature delivery through sprints and test-driven development (TDD).
- Contributed to culturally relevant health education for non-literate users, tailoring content for field testing.
- Contributed to weekly stand-ups and developer meetings with senior developers, improving the application's storyline and design while adhering to Agile methodologies.

Technologies: Dart, Flutter, Azure, Git, XML

Parallelizing Traveling Salesman Problem (TSP)

- Implemented a parallelized solution for the Traveling Salesman Problem (TSP) using Genetic Algorithms (GA) and OpenMP to reduce computational time.
- Achieved a 600% improvement in runtime by generating 50,000 trips across 150 generations for 36 cities using GA operations such as evaluation, selection, crossover, mutation, and population management.
- Deployed a Greedy Crossover Algorithm to ensure unique datasets and prevent revisiting cities, improving the accuracy of the results.

Technologies: Java, OpenMP

Parallelizing 2-D Wave Diffusion

- Developed a parallelized simulation for 2-D wave diffusion, utilizing complex physics formulas to model wave patterns, achieving a 200% reduction in calculation time.
- Integrated Schrödinger's wave equation into the simulation to calculate wave heights based on neighboring cells.
- Applied parallelization techniques using MPI and OpenMP, dividing the simulation space to optimize performance and ensure accurate calculations.
- Evaluated performance by comparing parallelized versions against sequential execution in various scenarios.

Technologies: C++, MPI, Hadoop